

MODEL QUESTION PAPER
Design of Machine Elements

Time :3Hours

Max. Marks :75

PART A

I Answer all questions in one word or one sentence

1	A combination of kinematic pairs, joined in such a way that the relative motion between the links is completely constrained, is stated as.....	M 1.02	R
2	List any two permanent and temporary fasteners.	M 1.03	R
3	List the general torsion equation for cylindrical shafts	M 2.01	R
4	Name the key which is likely to slip under the load	M 2.03	R
5	The point on the pitch curve at which the pressure angle is maximum is stated as the -----	M 3.01	R
6	In a machine ----- takes care of fluctuations of speed during a cycle	M 3.03	R
7	List any four power drives	M 4.01	R
8	The size of a gear is usually stated by A) Pressure angle (B) Pitch circle diameter (C) Circular pitch (D) Diametral pitch	M 4.04	R
9	Recall the equation for ratio of belt tensions in flat belt	M 4.02	R

(9x1=9 marks)

PART B

II Answer any **eight** questions from the following .Each question carries 3 marks.

1	Explain factor of safety for ductile materials and brittle materials	M 1.01	U
2	Identify any three desirable properties of shaft materials	M 2.01	U
3	Classify shafts	M 2.01	U
4	Derive an equation to find the diameter of solid shaft when the shaft is subjected to bending moment only	M 2.02	U

5	Explain woodruff key with figure	M 2.03	U
6	Classify cams	M 3.01	U
7	Explain turning moment diagram for single cylinder 4-Stroke I.C. Engine	M 3.03	U
8	List any three functions and requirements of shaft couplings	M 3.05	R
9	Draw the radial and thrust bearings and mark the elements	M 3.06	U
10	List the disadvantages of chain drive	M 4.03	R

(8x3=24 marks)

PART C

III Answer **ALL** questions. Each question carries 7 marks

1	Explain different types of constrained motion in the kinematic pairs	M 1.02	U
	OR		
2	A plate 100 mm wide and 10 mm thick is to be welded to another plate by means of double parallel fillets. The plates are subjected to a static load of 80 kN. Calculate the length of weld if the permissible shear stress in the weld does not exceed 55 MPa.	M 1.03	A
3	Explain double slider crank chain with first inversion	M 1.02	U
	OR		
4	Explain with figure single riveted lap joint and single riveted butt joint	M 1.04	U
5	Design the rectangular key for a shaft of 50 mm diameter. The shearing and crushing stresses for the key material are 42 MPa and 70 MPa.	M 2.03	A
	OR		
6	A line shaft rotating at 200 r.p.m. is to transmit 20 kW. The shaft may be assumed to be made of mild steel with an allowable shear stress of 42 MPa. Compute the diameter of the shaft, neglecting the bending moment on the shaft.	M 2.02	A
7	Construct the profile of a cam operating a knife edged follower from the following data: 1. Follower to move outwards through 35 mm during 60 degree of cam rotation 2. Follower to dwell for the next 40 degree 3. Follower to return to its original position during next 90 degree 4. Follower to dwell for the rest of the cam rotation The displacement of the follower is to take place with simple harmonic motion during both the outward and return stroke. Line of stroke of the follower is offset 10 mm from the axis of the camshaft. The least radius of the cam is 25 mm.	M 3.01	A

	OR		
8	With a sketch explain the working of a simple watt governor	M 3.02	U
9	List any seven advantages of V-belt drive over flat belt drive	M 4.01	R
	OR		
10	An engine running at 150 r.p.m. drives a line shaft by means of a belt. The engine pulley is 750 mm diameter and the pulley on the line shaft is 450 mm. A 900 mm diameter pulley on the line shaft drives a 150 mm diameter pulley keyed to a dynamo shaft. Calculate the speed of dynamo shaft, when(1). there is no slip, and (2). there is a slip of 2% at each drive.`	M 4.02	A
11	With a sketch explain the reverted gear train and describe it's velocity ratio.	M 4.04	U
	OR		
12	The following data relate to two meshing gears. Velocity ratio =1/3, m =4 mm. Centre distance =200 mm .Compute the number of teeth.	M 4.04	A

(6x7=42 marks)